

Loan Late Payment Reminder - Notification System Flow

Generated: 2026-04-24 03:41:19

1 ■ DAILY SCHEDULED PROCESSING

Component	Action	Details
Scheduler	Triggers processPendingSmsNotifications	Default: Daily at configured time
Data Source	SELECT from Oracle VIEW_LOAN_LATE_PAYMENT	LocalMockTestDataHelper (2 test records)
Record Count	Fetch 3,950+ loan late records	Production connects to Oracle DB
Grouping	Create batches of 20 records	Configured: app.notification.batch-size

2 ■ STORE TO QUEUE WITH JSON PAYLOAD

Step	Process	Storage
Build JSON	payloadBuilder.buildJsonPayload(phoneNo)	Stored in jsonPayload field
Create Record	NotificationQueue with all fields	status=PENDING, jsonPayload
Persist	notificationQueueRepository.save()	INSERT into notification_queue
Ready	Log: "Queued: X records..."	Ready for async processing

3 ■ ASYNC BATCH PROCESSING (Spring @Async)

Component	Configuration	Details
Executor	@Async("notificationExecutor")	Spring-managed thread pool
Pool Size	max-threads: 10	app.notification.max-threads
Transaction	@Transactional on method	Context propagated to async thread
Batch Size	processSingleBatchAsync()	Process 20 records per batch worker
Parallel	CompletableFuture.allOf()	Wait for all batches to complete

4■■ SEND TO NOTIFICATION API

Scenario	Action	Database Update
Local Profile	log.info("SMS dispatch (LOCAL)...")	Returns SUCCESS (no API call)
Production Success	POST /SendOTT with JSON payload	status=SUCCESS, processed_at=NOW()
Production Failure	Catch RestClientException	status=FAILURE, failure_reason, retry++
Retry Logic	Optimistic lock handler (3 attempts)	Exponential backoff: 50ms→100ms→150ms
DB Operation	Native UPDATE query (bypass ORM)	Avoids version conflict issues

5■■ LOGGING & AUDIT TRAIL

Table	Records	Purpose
notification_queue	Working records only	Operational - Auto-deleted after 2 days
notification_log	All deliveries + JSON payloads	Permanent audit trail - Keeps forever
notification_processed	Daily summary	Cache & reporting (total/success/failure)
Key Fields	jsonPayload, status, timestamps	Complete traceability of every SMS

6■■ DAILY AUTO-CLEANUP (2 AM CRON JOB)

Action	Condition	Benefit
Delete from notification_log	WHERE status=SUCCESS AND created_at < NOW() - 2 days	Keeps log lean & performant
Keep notification_log	All records preserved forever	Permanent compliance audit trail
Scheduled	@Scheduled(cron="0 0 2 * * *")	Daily 2 AM cleanup runs automatically
Result	Log: "Queue cleanup: Deleted X records"	Queue never grows indefinitely

■ SYSTEM FEATURES & OPTIMIZATIONS

- ✓ **Async Processing:** Spring @Async with transaction context propagation
- ✓ **Reliability:** Optimistic locking retry (3 attempts), Exponential backoff delays
- ✓ **Performance:** Batch processing (20/batch), Multi-threaded (10 max), Auto-cleanup
- ✓ **Auditing:** All JSON payloads stored, Permanent notification_log, Daily tracking
- ✓ **Constants:** Centralized NotificationConstants, No magic strings
- ✓ **Testing:** TestDataHelper for local mode, No database needed for testing
- ✓ **Soft Delete:** is_deleted flag for data retention compliance
- ✓ **Configuration:** Environment profiles (local/dev/prod), Config-driven settings